Sparse Autoencoders Recover Reproducible Structural Signal in Protein Language Model Latent Space Representations

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Generative protein-design models produce candidate binders without revealing what internally distinguishes a strong binder from a weak one, limiting rational improvement and failure diagnosis. We apply sparse autoencoders (SAEs), a mechanistic-interpretability technique, to residue-level ESM-C activations of Boltz-designed binder candidates against Vilip-1, a blood biomarker of acute neurological injury, to identify human-interpretable features in these latent representations without modifying any binder sequences. Mixing real, evolutionarily distinct natural-protein sequences into SAE training was the largest lever for generalization beyond synthetic designs alone, raising held-out natural-protein reconstruction fidelity from 0.39 to 0.60 fraction of variance explained. To test whether learned features reflect genuine protein structure rather than training artifacts, we compared two independently trained SAE dictionaries: binder-alone encoding versus encoding with full binder-target cross-attention before target positions were discarded. Despite different encodings, each dictionary's most generic features converge on the same real-protein residues at a rate chance cannot explain (hypergeometric test against the $\sim$7M-residue candidate space, $p \approx 7 \times 10^{-28}$), confirming that cross-attention preserves the learned signal. This convergence survives a shared-seed confound check (5 of 6 agreeing cases use different learned feature indices per dictionary) and is corroborated by InterPro domain annotations and calibrated LLM-assisted labeling that withholds a description when evidence is scattered. Together, these results show that SAE features recovered from protein language model activations carry real, reproducible structural signal, a necessary validation step before using such features to inform or steer binder design.